Electrical Fuse Schedule (Implementation - Lynx Topology)

As-of date: `2026-07-19`

Purpose: define each required fuse by circuit, protected conductor/device, holder/housing method, physical placement, and linked wire-gauge assumptions for the approved Phase 1 Lynx architecture with a battery-backed 12V bus and dedicated 48V secondary alternator branch.

Related docs:

- Canonical electrical/system baseline: `docs/core/SYSTEMS.md`
- Implementation topology and conductor map: `docs/implementation/ELECTRICAL_overview_diagram.md`
- Decisions/open items tracker: `docs/core/TRACKING.md`
- BOM source of truth: `bom/bom_estimated_items.csv`

Design Basis

- Topology: `Victron Lynx Distributor M10` (`LYN060102010`) with `4` active fused `48V` branches; Slot 4 feeds Orion input.
- Battery bank assumption: `3x 48V 100Ah` batteries in parallel (`3` separate battery-positive conductors leaving batteries).
- 12V distribution assumption: 12V fuse block used as the shared junction device (main `+` stud = source combine, integrated negative bus/main `-` = return), fed by Orion-Tr Smart `48/12-30` charger and a `12V 100Ah LiFePO4` buffer battery branch.
- Parallel-bank safety rule: use one Class T fuse per battery-positive conductor leaving the battery.
- Active branch devices on Lynx:

1. MultiPlus-II `48/3000`
2. SmartSolar `150/45`
3. Dedicated `48V` secondary alternator branch (Mechman/WS500 path)
4. Slot 4 / `F-05` — Orion-Tr `48/12-30` input through one `40A` MEGA body-marked at least `58VDC` ; standalone `F-06` is retired.

- Current shop identification note: do not infer the purpose of the physically X-marked/misrated Lynx fuse from the X alone. Identify its slot. Slot 2 requires `F-03 60A/80V` MEGA for the MPPT; Slot 4 requires `F-05 40A` MEGA marked `>=58VDC` for Orion. No `32V` -rated fuse is acceptable on an energized `48V` branch.
- Alternator path lock for this pass:

1. `F-04` is locked to `150A` MEGA (`58V/80V`) in Lynx Slot 3 at the house-bank/Lynx end of the alternator positive run.
2. Obsolete pre-Mechman charger/fuse paths are removed from active architecture and primary layout planning.
3. Confirmed `PH-VAN` harness uses one short red/black pair as combined regulator power and voltage sense at the house/main bus. Treat former separate `F-12` regulator-power and `F-13` positive-sense functions as one fused `PH-VAN` red lead for this install; current Wakespeed VAN diagram shows `15A` at that lead. Current-sense high/low remains an unfused twisted sense pair per Wakespeed manual.
4. Ford `Upfitter Switch #3` is locked as the manual `WS500` enable source through `F-15` .

Alternator Branch Fuse + Wire Finalization (`2026-03-19`)

Assumptions for this pass:

1. One-way alternator run length: `20 ft` .
2. Dedicated positive and dedicated negative runs (loop length `40 ft`).
3. Charging target voltage basis: `58.4V` .
4. Voltage-drop planning target: about `<=2%` under expected charging current.

Resistance and drop screen (`V_drop = I * (2 * L * R_per_ft)`):

Gauge	`150A` drop	`%` @ `58.4V`	`200A` drop	`%` @ `58.4V`	Decision note
`4 AWG`	`1.49V`	`2.55%`	`1.99V`	`3.40%`	Reject for this run length/current class
`2 AWG`	`0.94V`	`1.61%`	`1.25V`	`2.14%`	Acceptable for `~150A` class
`1/0 AWG`	`0.59V`	`1.01%`	`0.79V`	`1.35%`	Strong margin
`2/0 AWG`	`0.47V`	`0.80%`	`0.62V`	`1.07%`	Best margin; inventory already on hand

Lock for this build pass:

- Keep the planned alternator branch fuse at `150A` (`F-04` , Lynx Slot 3).
- Reuse existing uncut `2/0` inventory for alternator `+` and dedicated negative run.

- Inactive BOM row `173` preserves the removed contingency estimate; open a new active line only if field measurement proves an actual shortfall.

Required Fuse Map (Start-to-Finish, With Housing)

Fuse ID	Circuit (source -> load)	Protected wire/device	Fuse type and voltage class	Amperage	Holder or housing method	Physical location	Pl co g
F-01A	Battery A + -> bank positive combine/disconnect input	Battery A positive cable leaving battery	Class T (>=125VDC)	200A provisional	Blue Sea Class T fuse block (covered stud mount, 110A-200A family)	Battery compartment, within ~`7"` of Battery A positive post	2/0
F-01B	Battery B + -> bank positive combine/disconnect input	Battery B positive cable leaving battery	Class T (>=125VDC)	200A provisional	Blue Sea Class T fuse block (covered stud mount, 110A-200A family)	Battery compartment, within ~`7"` of Battery B positive post	2/0
F-01C	Battery C + -> bank positive combine/disconnect input	Battery C positive cable leaving battery	Class T (>=125VDC)	200A provisional	Blue Sea Class T fuse block (covered stud mount, 110A-200A family)	Battery compartment, within ~`7"` of Battery C positive post	2/0
F-02	Lynx Slot 1 -> MultiPlus DC+	Main inverter positive feeder	MEGA, 58V or 80V	125A	Integrated Lynx Distributor fuse slot	Lynx Distributor, Slot 1	2/0 pla (AW mi on rur
F-03	Lynx Slot 2 -> SmartSolar BAT+	MPPT battery-side positive feeder	MEGA, 80V Victron replacement stock	60A	Integrated Lynx Distributor fuse slot	Lynx Distributor, Slot 2	6 A
F-04	Lynx Slot 3 -> 48V alternator branch input (ALT+)	Alternator-to-house positive charge cable	MEGA, 58V or 80V	150A	Integrated Lynx Distributor fuse slot	Lynx Distributor, Slot 3	2/0 (re loc
F-05	Lynx Slot 4 -> Orion 48V input +	Orion 48V input feeder	MEGA 40A, body-marked >=58VDC; Victron CIP138040020 40A/80V replacement fallback	40A deliberate feeder-protection value; above Victron's 20A table recommendation but safely below installed 6 AWG ampacity	Integrated Lynx Distributor fuse slot	Lynx Distributor, Slot 4	Exi AWG to no inp
F-06	Retired standalone Orion input fuse position	Not active	N/A	Not installed	No holder	N/A	Pu 30A MII FKS

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F-07	Orion 12V output + -> 12V fuse block main + stud	Main 12V feeder from Orion into shared source-combine point	MEGA, 80V Victron replacement stock	60A	Victron MEGA fuse holder (external, non-Lynx)	Electrical cabinet, within ~`7"of Orion12V` output stud	6 A pla AWG mir per cab tab
F-09A/B/C	PV string + leads -> MPPT PV combiner	Each solar string positive conductor and reverse-current path	gPV string fuse (>=150VDC)	15A each (provisional)	10x38 touch-safe PV fuse holders in weatherproof combiner enclosure	Roof-entry combiner near gland/pass-through	10 wir
F-10	12V fuse block branch circuits -> each 12V load	Individual 12V branch conductors and load circuits	ATO/ATC blade fuses (32V class)	Per-circuit	Integrated sockets in marine 12V fuse block	12V fuse block in electrical cabinet	Per
AUDIO-HU / 12V-12	12V fuse block -> Kicker 46KMC2 media center/source unit	KMC2 source/head-unit branch conductor and device harness	ATO/ATC branch fuse (32V class) plus KMC2 harness 15A ATM fuse	15A	Integrated 12V fuse block branch plus OEM KMC2 harness fuse	Electrical cabinet to driver-side DC shelf/source face	12 dup kep arc ft; rou gro tow ft+
AUDIO-SUB	12V source/main + stud -> external fuse -> Kicker 49PTRTP10 powered sub +	Powered-sub positive branch and subwoofer input; manual-required external protection	Inline/MRBF/AFS/ANL-class fuse matched to selected 4 AWG kit (32V+ DC class acceptable on 12V branch)	40A	Holder within about 18 in of 12V source takeoff; if using Kicker 47KMPK4, fit the PTRTP10-required 40A fuse rather than the kit generic larger fuse	Near 12V source/junction, downstream of SW-12V-BATT	4 A pos wit ma AWG to neg bus stu
F-11	12V buffer battery + -> 12V fuse block main + stud via SW-12V-BATT	Buffer battery source cable and downstream junction fault exposure	Inline MIDI/AMI/ANL family rated >=32VDC	100A class baseline	Sealed inline holder mounted close to battery positive	Within ~`7"` of 12V buffer battery positive post	4 A pla

F-12/F-13-PHVAN	PH-VAN WS500 red lead at house/main positive bus	Combined WS500 regulator power and positive voltage-sense lead	Inline fuse/holder rated for actual 48V bank maximum; current Wakespeed VAN/internal-BMS diagram shows this fused at 15A	15A unless Wakespeed/Mechman profile guidance overrides	Sealed inline holder close to the house/main positive bus takeoff feeding the short PH-VAN red lead	Electrical panel near WS500/main bus; do not extend the short VAN red/black pair	Ha lea
CERBO-PWR	48V system positive -> Cerbo GX power +	Cerbo GX low-current electronics feed	Inline fuse/holder rated for 48V bank maximum voltage	1A-3A	Small inline holder close to the positive takeoff	Electrical cabinet, preferably system/load side of main disconnect so Cerbo powers down with the house system during bench shutdown	18 rec dup acc
F-15	Ford Upfitter #3 -> WS500 brown ignition/enable wire	Low-current regulator enable/control wire	Inline ATC/ATO (32V class acceptable; 12V control circuit)	3A	Sealed inline holder near the Ford upfitter blunt-cut source / splice handoff	Engine bay or control-wire handoff point before small-gauge run to WS500	16 TX
OEM-SHUNT	Battery-side positive or Lynx/system positive tap -> SmartShunt Vbatt+ terminal	SmartShunt electronics supply/sense lead	External Victron-supplied red cable with inline low-current fuse; not an internal SmartShunt fuse	OEM value	Inline holder in supplied red cable	Prefer battery-side positive if SOC continuity is desired while the main disconnect is open; system side is acceptable if zero parasitic draw during disconnect-off storage matters more	OE han lea

Retired Fuse IDs

Obsolete pre-Mechman charger/fuse paths are removed from the active schedule. Do not reserve board space or labels for them.

Spare Fuse Inventory (Updated)

Fuse type	Installed qty	Spare qty to carry	Notes
Class T 200A (provisional installed)	3	1 on hand	Owner confirmed 3 holders and 4 slow-blow Class T fuses total (3 installed + 1 spare); add more only if the one-spare-per-installed policy is later desired
MEGA 150A (58V/80V)	1	2	Alternator branch (F-04) installed + spare set (row 170)

MEGA 125A (58V/80V)	1	4	MultiPlus branch
MEGA 60A (80V)	2	3	Victron 5-pack row 188: install MPPT (F-03) + Orion output (F-07), keep 3 spares. Earlier 6x low-cost 60A MEGA batch was owner-confirmed misadvertised/not actually 58V and is quarantined/trash, not 48V install stock.
MEGA 40A, >=58VDC	1 active	1-2	Orion input in Lynx Slot 4 (F-05); existing body-marked 58VDC stock is acceptable under the locked 56.8V charge ceiling; use Victron CIP138040020 40A/80V as replacement fallback
Retired Orion standalone input-fuse stock	0 active	Optional purchased stock only	30A/58V MIDI and FKS/ATO parts are not installed; no USM1/ATM20 purchase required
12V buffer battery main fuse (100A class)	1	3	Spare pack basis is BOM row 105
WS500 PH-VAN combined regulator power / positive-sense fuse (F-12/F-13-PHVAN)	1 active position	2	Carry 15A spares with holder/fuse voltage rating verified for the 48V bank maximum
Cerbo GX power fuse (CERBO-PWR)	1 active position	1	Carry a spare 1A-3A fuse/holder rated for the 48V bank maximum
WS500 ignition/enable fuse (F-15)	1 active position	2	Carry spare 3A mini/ATO fuse and one spare sealed holder; 12V control circuit
PV string fuse 15A gPV	3	3	One spare per string
SmartShunt OEM harness fuse	1	1	Keep OEM-equivalent spare if field-replaceable
ATO/ATC branch fuses	variable	2 each used value	Keep mixed kit onboard
Camper audio KMC2 branch (AUDIO-HU)	1 active 15A branch plus KMC2 harness 15A ATM	2 spare 15A blade/ATM fuses	Preserve both source-side conductor protection and the KMC2 harness fuse
Camper audio powered sub (AUDIO-SUB)	1 active 40A external fuse	1-2 spare 40A fuses matching the selected holder family	PTRTP10 manual calls for 40A external fuse and 4 AWG power/ground

BOM Row Mapping

Fuse scope	BOM row(s)
Main battery Class T protection (F-01A/F-01B/F-01C) + Class T spares	bom/bom_estimated_items.csv row 7
Lynx branch MEGA fuses (F-02 to F-05 installed) + spare set	bom/bom_estimated_items.csv rows 10, 170, 188, and 323
Orion installed fuse-holder hardware (F-05 Lynx input, F-07 external output)	bom/bom_estimated_items.csv rows 6 and 11
Retired Orion standalone input-fuse stock (F-06)	Purchased stock: active BOM rows 133, 182, and 321; retired holder/bridge history: inactive BOM rows 326 and 230
WS500 regulator-power and voltage-sense protection (F-12/F-13-PHVAN)	bom/bom_estimated_items.csv rows 171 and 320

Cerbo GX power feed (CERBO-PWR)	bom/bom_estimated_items.csv row 22; small inline fuse/holder may come from low-current install stock
WS500 Upfitter #3 enable/control path (F-15)	bom/bom_estimated_items.csv row 176
12V buffer battery (B12)	bom/bom_estimated_items.csv row 21
12V buffer battery main fuse + holder (F-11)	bom/bom_estimated_items.csv row 125
12V battery disconnect (SW-12V-BATT)	bom/bom_estimated_items.csv row 124
12V branch panel and blade fuses (F-10)	bom/bom_estimated_items.csv row 16
Camper audio source/head-unit branch (AUDIO-HU / 12V-12)	bom/bom_estimated_items.csv rows 189, 192, and 193
Camper audio powered-sub branch (AUDIO-SUB)	bom/bom_estimated_items.csv rows 191 and 192
SmartShunt external OEM fused red lead (OEM-SHUNT)	bom/bom_estimated_items.csv row 23 (included with SmartShunt kit)
PV string fuses + holder (F-09A/B/C) and spares	bom/bom_estimated_items.csv row 106

Assumptions and Open Items

1. Wire sizing above assumes copper conductors and enclosed vehicle routing.
2. `F-09A/B/C` and the old `3S3P` fuse count are modeling placeholders only; final PV fusing waits until solar modules/stringing are selected after shore and alternator charging are working.
3. Confirm WS500 `PH-VAN` harness fuse-holder voltage rating before final install; current build default is one `15A` fused red lead at the house/main positive bus for combined regulator power and positive voltage sense.
4. `SW-12V-BATT` remains manual-only in Phase 1; no automatic LVD behavior is assumed.
5. Final lock for `F-11` still requires explicit 12V buffer battery/BMS continuous discharge-current confirmation.
6. `F-15` exists to protect the smaller-gauge control wire between Ford `Upfitter #3` and the WS500 brown ignition/enable wire; the factory `25A` upfitter circuit protection is not the final wire-protection device for that branch.
7. Orion `F-05/F-06` lock: the active topology uses `F-05`, one `40A` MEGA body-marked at least `58VDC` in Lynx Slot 4, feeding the existing `6 AWG` Orion input directly. `F-06` is retired. This deliberately prioritizes simple feeder protection over strict adherence to Victron's `20A` table recommendation; do not stack an inline fuse after the Lynx fuse.
8. Camper audio `12V` load: KMC2 source branch and PTRTP10 powered-sub branch are downstream 12V loads, not new 48V branches. The PTRTP10 `40A` branch can exceed Orion `30A` continuous output during heavy bass; rely on the 12V buffer battery for transients and validate sustained loud-use behavior before assuming all 12V loads can run at max simultaneously.
9. Orion holder correction: no standalone input holder is required. The purchased `178.6150.0001` housing-only parts and `30A/58V` MIDI stock remain unused; do not salvage or stack them into the final branch. Torque the Slot 4 MEGA hardware and Orion terminals to current manufacturer values, tug-test, and strain-relieve the existing `6 AWG` pair.